FOR REPRESENTABLE MORPHISMS, LIMIT PRESERVATION ON OBJECTS EQUALS LOCAL FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Characterization of locally finite presentation. spaces-limits-proposition-characterize-locally-finite-presentation (cross-chapter, named). Descends morphisms along directed limits of affines in the (2)=>(1) direction.
- Surjection criterion. spaces-limits-lemma-surjection-is-enough (cross-chapter, named). Yields local finite presentation from surjectivity of the colimit map in the (1)=>(2) direction.

Source reference. Original reference 03YK: focused-seeds/geometry/S9-crit-representable-by-spaces-limit-preserving/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CM6: focused-seeds/geometry/S9-crit-representable-by-spaces-limit-preserving/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04AK: focused-seeds/geometry/S9-crit-representable-by-spaces-limit-preserving/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05N0: focused-seeds/geometry/S9-crit-representable-by-spaces-limit-preserving/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $p : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. If p is representable by algebraic spaces, then the following are equivalent:*

(1) *p is limit preserving on objects, and*
(2) *p is locally of finite presentation (see Algebraic Stacks, Definition 03YK).*

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Proof. Assume (2). Let $U = \lim_{i \in I} U_i$ be the directed limit of affine schemes U_i over S, let y_i be an object of $\mathcal{Y}$ over U_i for some i, let x be an object of $\mathcal{X}$ over U, and let $\gamma : p(x) \to y_i|_U$ be an isomorphism. Let X_{y_i} denote an algebraic space over U_i representing the 2-fibre product

$$(Sch/U_i)_{fppf} \times_{y_i, \mathcal{Y}, p} \mathcal{X}.$$

Note that $\xi = (U, U \to U_i, x, \gamma^{-1})$ defines an object of this 2-fibre product over U. Via the 2-Yoneda lemma ξ corresponds to a morphism $f_\xi : U \to X_{y_i}$ over U_i. By Limits of Spaces, Proposition 04AK there exists an $i' \geq i$ and a morphism $f_{i'} : U_{i'} \to X_{y_i}$ such that f_ξ is the composition of $f_{i'}$ and the projection morphism $U \to U_{i'}$. Also, the 2-Yoneda lemma tells us that $f_{i'}$ corresponds to an object $\xi_{i'} = (U_{i'}, U_{i'} \to U_i, x_{i'}, \alpha)$ of the displayed 2-fibre product over $U_{i'}$ whose restriction to U recovers ξ. In particular we obtain an isomorphism $\gamma : x_{i'}|U \to x$. Note that $\alpha : y_i|_{U_{i'}} \to p(x_{i'})$. Hence we see that taking $x_{i'}$, the isomorphism $\gamma : x_{i'}|U \to x$, and the isomorphism $\beta = \alpha^{-1} : p(x_{i'}) \to y_i|_{U_{i'}}$ is a solution to the problem.

Assume (1). Choose a scheme T and a 1-morphism $y : (Sch/T)_{fppf} \to \mathcal{Y}$. Let X_y be an algebraic space over T representing the 2-fibre product $(Sch/T)_{fppf} \times_{y, \mathcal{Y}, p} \mathcal{X}$. We have to show that $X_y \to T$ is locally of finite presentation. To do this we will use the criterion in Limits of Spaces, Remark 05N0. Consider an affine scheme $U = \lim_{i \in I} U_i$ written as the directed limit of affine schemes over T. Pick any $i \in I$ and set $y_i = y|_{U_i}$. Also denote i' an element of I which is bigger than or equal to i. By the 2-Yoneda lemma morphisms $U \to X_y$ over T correspond bijectively to isomorphism classes of pairs (x, α) where x is an object of $\mathcal{X}$ over U and $\alpha : y|_U \to p(x)$ is an isomorphism. Of course giving α is, up to an inverse, the same thing as giving an isomorphism $\gamma : p(x) \to y_i|_U$. Similarly for morphisms $U_{i'} \to X_y$ over T. Hence (1) guarantees that the canonical map

$$\operatorname{colim}_{i' \geq i} X_y(U_{i'}) \longrightarrow X_y(U)$$

is surjective in this situation. It follows from Limits of Spaces, Lemma 0CM6 that $X_y \to T$ is locally of finite presentation. □